

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12418164
Kind Code	B2
Date of Patent	September 16, 2025
Inventor(s)	White; Thomas C.

Non-welded vertical cable tray redirector

Abstract

Vertical redirector components speed assembly and connection of cable trays. More efficiently manufactured, universal components needed throughout vertical redirectors of a specific size and type cable tray run reduce the overall total items and cost of inventory. The improvement is in part the elimination of welding cable redirector components in favor of connecting the components together with bolts and nuts in the field. The inventory and shipment of individual parts is substantially more efficient because the new components “nest” efficiently prior to assembly. Moreover, components are designed to minimize the number of bolts and nuts used without jeopardizing strength. The use of bolts and nuts to fasten the components also enables good electrical conductivity for effective electrical grounding of the assembled cable tray system.

Inventors:	White; Thomas C. (Greenville, SC)
Applicant:	James C. White Company, Inc. (Greenville, SC)
Family ID:	1000008818576
Assignee:	James C. White Company, Inc. (Greenville, SC)
Appl. No.:	18/522747
Filed:	November 29, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240106213 A1	Mar. 28, 2024

Related U.S. Application Data

continuation parent-doc US 16892974 20200604 US 11171470 20211109 child-doc US 17498924
continuation-in-part parent-doc US 17498924 20211012 US 11936172 child-doc US 18522747

Publication Classification

Int. Cl.: H02G3/04 (20060101); F16L3/26 (20060101); G02B6/44 (20060101)

U.S. Cl.:

CPC H02G3/0456 (20130101); F16L3/26 (20130101); G02B6/4453 (20130101)

Field of Classification Search

CPC: H02G (3/0456); F16L (3/26); G02B (6/4453)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5469893	12/1994	Caveney et al.	N/A	N/A
8167250	12/2011	White	248/68.1	H02G 3/0608
9184578	12/2014	Winn et al.	N/A	N/A
9841123	12/2016	White	N/A	N/A
11171470	12/2020	White	N/A	N/A
11936172	12/2023	White	N/A	N/A
2012/0312761	12/2011	White	N/A	N/A
2016/0077299	12/2015	Tally	385/135	G02B 6/4459

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
208433706	12/2018	CN	N/A
102017120728	12/2018	DE	N/A
1249653	12/2001	EP	N/A
1249653	12/2003	EP	N/A
2211436	12/2009	EP	H02G 3/0437
2682311	12/1992	FR	N/A
2267605	12/1992	GB	H02G 3/0608
2314216	12/1996	GB	F16L 3/26
2527600	12/2018	GB	N/A
191621	12/2005	PL	N/A
9307665	12/1992	WO	N/A
2012127087	12/2011	WO	N/A
WO-2017147256	12/2016	WO	H02G 3/0608

OTHER PUBLICATIONS

James White Tubetrack 2017 product catalog,
https://fwwebbimage.fwebb.com/ProductInfo/tubetrack_online.pdf (Year: 2017). cited by examiner

Click and Go youtube video titled “Click and Go!—Instructive #7/ Using Vertical Bend” dated Sep. 17, 2013, <https://www.youtube.com/watch?v=kIZCk-IVAU0> (Year: 2013). cited by examiner

Legrand Cablofil tough tray product catalog dated Jun. 23, 2016 <https://legrand.cr/wp-content/uploads/2023/05/cablofil-toughtray-2017-18.pdf> (Year: 2016). cited by examiner

Vergokan snap-in rising elbow SBCL online product page, <https://www.vergokan.com/en/cable-support-systems/cable-trays/Accessories/SBCL> dated on archive.org at Apr. 14, 2019 (Year: 2019). cited by examiner

Obo-bettermann vertical 90 degrees ascending RBV sold on amazon.com first available date: Feb. 26, 2015 <https://www.amazon.de/-/en/obo-bettermann-Curved-Vertical-Ascending-s-ft/dp/B00B7ZJCT2> (Year: 2015). cited by examiner

Cable-Race® dedicated tray support system catalog from James C. White Company, Inc, 29 pages, dated Aug. 2019 (Year: 2019). cited by applicant

James White Tubetrack 2017 product catalog for continuous tray support system (Year: 2017). cited by applicant

Techline Mfg. snap track catalog for Aluminum Cable Tray Systems, 76 pages, 2017 (Year: 2017). cited by applicant

Primary Examiner: McKinnon; Terrell L

Assistant Examiner: Tan; Ding Y

Attorney, Agent or Firm: Maynard Nexsen PC

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation-in-part application of U.S. patent application Ser. No. 17/498,924, and entitled “A Non-Welded Horizontal Cable Tray Redirector”, filed Oct. 12, 2021, which is a continuation application of U.S. Pat. No. 11,171,470, filed Jun. 4, 2020, and entitled “A Non-Welded Horizontal Cable Tray Redirector.” Accordingly, this application claims benefit of U.S. Pat. No. 11,171,470 and U.S. patent application Ser. No. 17/498,924 under 35 U.S.C. § 120. U.S. Pat. No. 11,171,470 and U.S. patent application Ser. No. 17/498,924 are incorporated herein in their entirety.

TECHNOLOGY FIELD

(1) This disclosure relates to cable trays in general and, in particular, to components of cable trays that change the direction, or “redirect” a cable tray. An example of a cable tray redirector is found in U.S. Pat. No. 8,573,409, which is incorporated herein in its entirety by reference.

BACKGROUND

(2) Cable trays are devices that support electrical cables. The term “cables” is intended broadly to include fiber optic cables, pipes, conduit, wiring, tubing, and the like, which may deliver electrical and optical signals, electricity, fluids, and data from one location to another. Cable trays can make it easier to install cables, to facilitate the connecting of cables, to locate and trace cables, and to replace, add or remove cables.

(3) Plural cable tray components may be connected together to form runs. The runs are usually elevated so that the cables they carry are out of harm's way, and not an obstruction to people or vehicles and are well above standing water and stored items. It is often easier to install and trace cables when they are in cable tray runs that are overhead than when they are on the ground where they compete for floor space with vehicles, inventory, finished goods, and pedestrian pathways.

(4) In many facilities, such as processing plants that include devices for instrument control and

monitoring, cables can be so numerous that several tray runs may be required, some oriented in different directions and at different elevations, perhaps changing direction repeatedly in order to avoid walls, columns and other cable tray runs when proceeding to their various destinations in the facility.

(5) Cable trays thus serve important purposes, particularly in plants requiring highly automated systems for monitoring and control such as, for example, office buildings, factories, warehouses, nuclear facilities, power plants and refineries.

(6) Thus, cable trays and the components used with cable trays need to enable designers to create cable tray runs that can reach their destination without interfering with other structures, and with the ability to change direction as needed in order to get them to where they are intended to go.

(7) A cable tray redirector, or simply a “redirector”, as used herein, refers to a component of a cable tray run that changes the direction of the cable tray run in a horizontal plane. A redirector may enable some of the cables in a tray to change direction by 90 degrees, for example, as some cables in the same cable tray run may continue forward in the original direction while others are directed in yet a third direction.

(8) More easily assembled, robust in operation, and economical cable tray redirectors would be an advantage in cable tray system design, construction and use. Better redirectors might have little or no welding requirements, use the same connecting fasteners as are commonly used to connect redirecting connectors to tray sections, and might be made of a metal that could be easily electrically grounded, not be bulky or difficult to store, package and ship, and could be used in a wide variety of cable tray applications.

SUMMARY

(9) According to its major aspects and briefly recited, herein is disclosed a set of components that are used to assemble cable tray redirectors that are useful for redirecting cable tray runs. The components are connected using the same bolts and nuts throughout, and which are used in the present redirector kit to avoid welded joints or expensive manufacturing processes. Welding takes time and may be unsuitable in some facilities, such as nuclear power plants and process facilities or where there are seismic restrictions because of concerns for stress and corrosion. The components of the present redirectors are made of materials that preserve electrical conductivity throughout any cable tray run, which makes electrical grounding easier. Also, the present cable tray redirectors enable cable redirection to the left and to the right, or both, up to 90 degrees in a horizontal plane. In addition, unlike prior art redirectors, the major components of the present system are designed to “nest”, which means that one component fits close to another of the same kind so that storage and shipping volumes are greatly reduced, requiring less warehouse, packaging or onsite space to maintain inventory. The fasteners used to assemble components are all interchangeable, too, so minimal numbers of part types need to be maintained.

(10) Current or known cable tray redirectors are fabricated as a weldment or a press formed metal, fiberglass or plastic product. The present redirectors reduce costs by eliminating welding time and excessive costs of press-forming and drawing tools. The present redirector is manufactured and assembled without welding.

(11) A feature of the present disclosure is a redirector that connects mechanically to straight tray sections. The present redirector requires no welding for connection to cable tray sections. The present redirector comprises multiple components, which are assembled using the same fasteners that connect the redirector kit to straight tray sections, using fasteners common to cable tray construction. In fact, the same number and type of fasteners required to connect a prior art welded redirector to cable trays is required for the assembly of the present redirecting kit to cable trays. Welding is eliminated without adding the cost of material and time to assemble the present redirectors to cable trays.

(12) Those skilled in the construction and operation of cable trays and redirectors will appreciate

these and other features and their advantages from a careful reading of the Detailed Description, accompanied by the following drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a prior art cable tray redirector for redirecting the horizontal course of a cable tray run by 90-degrees;
- (2) FIG. 2 is a perspective view of another prior art cable tray redirector for redirecting the horizontal course of a cable tray run by 45-degrees;
- (3) FIGS. 3A, 3B, and 3C, respectively, show perspective views of sections of a wide prior art cable channel tray (FIG. 3A), a narrow prior art cable channel tray (FIG. 3B); and a prior art L-shaped cable tray having one wall (FIG. 3C), wherein each tray has at least one row of holes for mounting clamps and guides to control the cables in the bed of the respective tray and for connecting tray sections and redirectors;
- (4) FIG. 3D shows a perspective view of a prior art vertical tray redirector that is press formed into shape;
- (5) FIG. 3E shows a perspective view of a prior art welded vertical tray redirector;
- (6) FIG. 4 is a perspective view of a vertical redirector base, according to an aspect of the disclosure;
- (7) FIG. 5 is a perspective view of a flange fitting, according to an aspect of the disclosure;
- (8) FIG. 6A shows an exploded, perspective view of a vertical redirector with two flange fittings in position to be joined to the vertical redirector base, according to an aspect of the disclosure;
- (9) FIG. 6B shows an assembled, perspective view of the vertical redirector of FIG. 6A with two flange fittings engaged to the vertical redirector base, according to an aspect of the disclosure;
- (10) FIG. 6C shows an assembled, back perspective view of the vertical redirector of FIGS. 6A and 6B with two flange fittings engaged to the vertical redirector base, according to an aspect of the disclosure;
- (11) FIG. 7 shows a perspective view of a vertical redirector base, according to an aspect of the disclosure;
- (12) FIG. 8 shows a perspective view of a vertical redirector flange fitting, according to an aspect of the disclosure;
- (13) FIG. 9A shows an exploded, perspective view of a vertical redirector with two flange fittings in position to be joined to the vertical redirector base, according to an aspect of the disclosure;
- (14) FIG. 9B shows an assembled, perspective view of the vertical redirector of FIG. 9A with two flange fittings engaged to the vertical redirector base, according to an aspect of the disclosure;
- (15) FIG. 9C shows an assembled, perspective view of the vertical redirector of FIGS. 9A and 9B with two flange fittings engaged to the vertical redirector base and two cable trays attached, according to an aspect of the disclosure;
- (16) FIG. 9D shows an assembled, back perspective view of the vertical redirector of FIGS. 9A, 9B, and 9C with two flange fittings engaged to the vertical redirector base and two cable trays attached, according to an aspect of the disclosure;
- (17) FIG. 10 is a perspective view of a nested stack of vertical redirector bases, according to an aspect of the disclosure;
- (18) FIG. 11 is a perspective view of a nested stack of flange fittings, according to an aspect of the disclosure;
- (19) FIG. 12 is a perspective view of a nested stack of vertical redirector bases adjacent a nested stack of flange fittings, according to an aspect of the disclosure;
- (20) FIG. 13A is a perspective view of a vertical redirector base, according to an aspect of the

disclosure;

(21) FIG. 13B is a perspective view of a flange fitting, according to an aspect of the disclosure;

(22) FIG. 14A shows an exploded, perspective view of a vertical redirector with two flange fittings in position to be joined to the vertical redirector base, according to an aspect of the disclosure;

(23) FIG. 14B shows an assembled, perspective view of the vertical redirector of FIG. 14A with two flange fittings engaged to the vertical redirector base and two cable trays, according to an aspect of the disclosure;

(24) FIG. 14C shows an assembled, back perspective view of the vertical redirector of FIGS. 14A and 14B with two flange fittings engaged to the vertical redirector base and two cable trays, according to an aspect of the disclosure;

(25) FIG. 15A is a perspective view of a flange fitting, according to an aspect of the disclosure;

(26) FIG. 15B is a perspective view of a vertical redirector base, according to an aspect of the disclosure;

(27) FIG. 15C is a perspective view of another flange fitting used in conjunction with the flange fitting of FIG. 15A, according to an aspect of the disclosure;

(28) FIG. 15D shows an exploded, perspective view of a vertical redirector with two flange fittings in position to be joined to the vertical redirector base and two cable trays, according to an aspect of the disclosure;

(29) FIG. 15E shows an assembled, back perspective view of the vertical redirector of FIG. 15D with two flange fittings engaged to the vertical redirector base and two cable trays, according to an aspect of the disclosure;

(30) FIG. 16A is a perspective view of a vertical redirector base, according to an aspect of the disclosure;

(31) FIG. 16B is a perspective view of a flange fitting, according to an aspect of the disclosure;

(32) FIG. 16C shows an exploded, perspective view of a vertical redirector with two flange fittings in position to be joined to the vertical redirector base and two cable trays, according to an aspect of the disclosure; and

(33) FIG. 16D shows an assembled, perspective view of the vertical redirector of FIG. 16C with two flange fittings engaged to the vertical redirector base and two cable trays, according to an aspect of the disclosure.

DETAILED DESCRIPTION

(34) Referring now to the drawings, FIGS. 1 and 2 show two prior art cable tray redirectors, and FIGS. 3A, 3B, and 3C show prior art cable trays that can be attached to the prior art cable tray redirectors of FIGS. 1, 2.

(35) FIG. 1, in particular, shows a redirector 20 with a curved (90-degree arc) wall 24. FIG. 2 shows a redirector 28 with a curved (45-degree arc) wall 32. Both 90-degree redirector 20 and 45-degree redirector 28 have welded corners 22, 30, respectively, and welded inside flanges 24, 32.

(36) Cable trays of various widths, such as those shown in the three prior art examples of FIGS. 3A, 3B, and 3C, can be attached to either end of a compatible 90-degree redirector or 45-degree redirector using the required number of bolts per cable tray. Wide cable trays such as the one shown in FIG. 3A can be attached to either end of 90-degree redirector 20 or 45-degree redirector 28 using four bolts per cable tray as required. The choice of cable tray will depend on the number and sizes of cables the tray is intended to carry. For example, a wide tray 36 as shown in FIG. 3A would be used to hold several cables or larger cables, a narrow tray 40 as shown in FIG. 3B would be selected when there are fewer cables or smaller cables to be carried. The cable tray 44 in FIG. 3C is for carrying a single cable or guided or clamped cables, bundles or tubes, and the like. FIGS. 3D and 3E show perspective views of prior art vertical tray redirectors (10, 11).

(37) The term “cable” will be used herein to indicate anything carried by a cable tray; including wiring, electrical cables, telephone cables, pipes for fluid delivery, pneumatic tubing, optical fibers, optical fiber bundles, and the like. The term “tray” or “cable tray” refers to a metal tray with a floor

and a wall on at least one side of the floor. The tray serves as a physical support and confinement for at least one cable running from one end of the tray to the other.

(38) Redirectors **20** and **28** and cable trays **36**, **40**, and **44** may have round, square, slotted, or a combination of round, square, and slotted holes formed therein to received various types of fasteners (not shown in FIGS. 1-5) that keep cables in place such as clamps or guides and for connecting tray sections and redirectors.

(39) An example of a vertical redirector base **80**, according to the present disclosure, is shown in perspective in FIG. 4. Vertical redirector base **80** includes a first end **82**, a second end **84**, wherein the first end **82** and the second end **84** are spaced apart from each other, a deck **86**, connecting the first end **82** and the second end **84**, and a pair of opposed tabs (**88a**, **88b**). The first end **82** and the second end **84** contain a front surface, a back surface, and an outer end. The deck **86** contains a front surface, a back surface, and a pair of opposed tabs (**88a**, **88b**) on the sides of the deck **86**. The deck **86** is preferably curved as illustrated in FIG. 4. An optional first transition **90** is defined by the intersection of the deck **86** and the first end **82**, and a second transition **92** is defined by the intersection of the deck **86** and the second end **84**. The first transition **90** serves as a transition portion between the deck **86** and the first end **82** that is offset from the deck **86**. The offset at the first transition **90** between deck **86** and first end **82** provides a smooth transition and can receive portion of a cable tray **36** placed adjacent the first end **82**, reducing the probability of cable damage by providing a smooth transition between the connected cable tray **36** and vertical redirector base **80**.

(40) Likewise, an optional second transition **92** serves as a transition portion between the deck **86** and the second end **84** that is offset from the deck **86**. The offset at the second transition **92** between deck **86** and second end **84** provides a smooth transition and can receive a portion of a cable tray **36** placed adjacent the second end **84**, reducing the probability of cable damage by providing a smooth transition between the connected cable tray **36** and vertical redirector **78**.

(41) The tabs (**88a**, **88b**) of deck **86** engage flange fittings **100**, shown in FIG. 5. FIG. 5 is a perspective view of a flange fitting **100**. The flange fitting **100** facilitates a cable redirection of about 90 degrees, for the sake of the present example. Those of ordinary skill will readily recognize that other angles for redirection are readily obtainable.

(42) Referring specifically to FIGS. 4 and 5, the deck **86** of vertical redirector base **80**, of FIG. 4, may receive a curved flange fitting **100**, of FIG. 5, against tab **88a** and tab **88b**. Flange fitting **100** includes a curved portion **102**, a first end **104** and a second end **106**. First end **104** carries a first flange **108** with at least one hole **110** formed therein. Second end **106** carries a second flange **112** with at least one hole **114**. Holes **110**, **114**, that are provided for attaching flange fitting **100** against tabs (**88a**, **88b**) of deck **86**, are preferably slotted holes to provide flexibility for connecting industrial components together.

(43) Also, referring to FIGS. 6A and 6B, the first flange **108** and second flange **112** of the flange fitting **100** elevates the bottom **116** of curved portion **102** so it clears the top of the tab **88a** of deck **86** when flange fitting **100** is being attached to deck **86**. The elevation of the curved portion **102** of each flange fitting **100** enables the curved portion **102** to fit above the tab **88b** of the deck **86** and is a feature of the present disclosure. The flange fitting **100** contains a tooth **130** extending outwardly from the curved portion **102** adjacent the first flange **108** and a tooth **130** extending outwardly from the curved portion **102** adjacent the second flange **112**. Recesses **132** are disposed within the curved portion **102**, extending from the outer edge of the curved portion **102** and into the curved portion **102** between the tooth **130** and the first flange **108** and the second flange **112** and having a u-shape. Likewise, flange fitting **100'** contains a tooth **130'** extending outwardly from the curved portion **102'** adjacent the second flange **112'** and a tooth **130'** extending outwardly from the curved portion **102'** adjacent the first flange **108'**. Recesses **132'** are disposed within the curved portion **102'**, extending from the outer edge of the curved portion **102'** and into the curved portion **102'** between the tooth **130'** and the first flange **108'** and the second flange **112'** and having a u-shape.

(44) Each tab (**88a**, **88b**) contains an outer edge **134** that extends to two opposed, spaced-apart shoulders **136**. The shoulders **136** extend from an outer edge of the deck **86** to the outer edge **134** of the tab (**88a**, **88b**) and on opposite ends of the deck **86**. A first shoulder **136** is disposed adjacent the first end **82** and the second shoulder **136** is disposed adjacent the second end **84**. As illustrated in FIG. 4, the shoulders **136** extend from the outer edge of the deck **86** at a 90° angle outwardly from the outer edge of the deck **86**. A corner **138** is formed at the intersection of the shoulders **136** and the outer edge of the deck **86**. While the shoulders **136** are adjacent the first end **82** and the second end **84**, respectively, the shoulders **136** are spaced apart from the first end **82**, first transition **90**, second end **84**, and second transition **92**, as illustrated in FIG. 4. A securing member **140** extends outwardly from both of the two opposed sides of the first end **82** and the second end **84**, as also illustrated in FIG. 4.

(45) FIGS. 6A and 6B show perspective views to illustrate moving flange fittings **100** into position onto deck **86** to form a vertical redirector **78** of the present disclosure. Flange fittings **100** are moved toward each side of the deck **86**, with tab **88a** centered so it can slide between first flange **108** and second flange **112** of one flange fitting **100**, and tab **88b** centered so it can slide between first flange **108'** and second flange **112'** of the second flange fitting **100'** on the opposite side of the deck **86**. First flange **108** and second flange **112** slide under deck **86** as curved portion **102** slides over tab **88a** and is positioned so that holes **94**, **96**, in first end **82** and second end **84**, respectively, are aligned with at least one hole **114** in first flange **112** or at least one hole **110** in second flange **108**, respectively, of the one flange fitting. Similarly, first flange **108'** and second flange **112'** slide under deck **86** as curved portion **102'** slides over tab **88b** and is positioned so that holes **94**, **96**, in first end **82** and second end **84**, respectively, are aligned with at least one hole **110'** in first flange **108'** or at least one hole **114'** in the second flange fitting **112'**. The teeth **130** and **130'** press against respective securing members **140** establishing the inside width between flange fittings **100** and **100'**. The recesses **132** and **132'** provide clearances necessary for forming the flange fitting and assembly of the vertical redirector **78**. The flange fitting **100** and **100'** does not extend beyond the first end **82** and the second end **84** of the vertical redirector base **80** or the outer edge of the receiving portion of the first end **82** or the second end **84**.

(46) FIG. 6B shows a completely assembled vertical redirector **78**. All of the components: vertical redirector base **80**, flange fittings **100** and **100'**, eight sets of fasteners (bolts **142** and nuts **144**) comprise a vertical redirector kit for connecting two cable trays **36** (not included as part of the kit). The components within the kit may be housed within a container, such as a bag, box, bin, or the like for storage, shipment, and containing all components within the kit together until assembly and use.

(47) It should be noted that in FIGS. 6A and 6B a total of eight bolts and nuts are needed to hold two cable trays within the non-welded cable tray redirector. FIG. 6B shows the same number: eight bolts and nuts, used to hold two cable trays to the vertical redirector **78** without any welding needed to complete the vertical redirector components or assembly. The cost and time to weld cable tray redirectors has been completely avoided. Cables (not shown) may be placed onto vertical redirector **78** beginning at first cable tray **36**, then across deck **86** and onto second cable tray **36'**. First cable tray **36** and second cable tray **36'** are secured to redirector base **80** using bolts **142** entering from the inside of first and second cable trays **36** and **36'**, respectively, and being fastened with nuts **144** from below the vertical redirector **78**.

(48) FIGS. 7, 8, 9A, 9B, and 9C illustrate another exemplary embodiment of the vertical redirector **78** and its component parts that contain all of the structures of the vertical redirector **78** illustrated in FIGS. 4, 5, 6A, and 6B and described above. The deck **86** in this embodiment may also contain two spaced-apart notches **146** disposed within each tab (**88a**, **88b**), as shown in FIG. 7. Each notch **146** is preferably disposed within the center of each tab (**88a**, **88b**) having two opposed side portions extending from the upper edge of the tab (**88a**, **88b**) to a bottom edge between each side portion. A securing member **140** extends outwardly from both of the two opposed sides of the first

end **82** and the second end **84**, as also illustrated in FIGS. **4** and **6A**. The flange fitting **100** contains a finger **148** extending from the curved portion **102**, as illustrated in FIG. **8**. The finger **148** extends outwardly from the bottom **116** of the curved portion **102** then perpendicularly to the curved portion **102** and is parabolic shaped having a conical upper portion.

(49) FIGS. **9A**, **9B**, and **9C** show perspective views to illustrate moving flange fittings **100** and **100'** into position onto deck **86** to form a vertical redirector **78** of the present disclosure. Flange fittings **100** are moved toward each side of the deck **86**, with tab **88a** centered so it can slide between first flange **108** and second flange **112** of one flange fitting **100**, and tab **88b** centered so it can slide between first flange **108'** and second flange **112'** of the second flange fitting **100'** on the opposite side of the deck **86**. First flange **108**, second flange **112**, and finger **148** slide under deck **86** as curved portion **102** slides over tab **88a** and finger **148** slides under notch **146**. The flange fitting **100** is positioned so that holes **94**, **96**, in first end **82** and second end **84**, respectively, are aligned with at least one hole **110** in first flange **108** or at least one hole **114** in second flange **112**, respectively, of the one flange fitting. Similarly, first flange **108'**, second flange **112'**, and finger **148'** slide under deck **86** as curved portion **102'** slides over tab **88b** and finger **148'** slides under notch **146**. The flange fitting **100'** is positioned so that holes **94**, **96**, in first end **82** and second end **84**, respectively are aligned with at least one hole **110'** in first flange **108'** or at least one hole **114'** in second flange **112'**, respectively, of the second flange fitting **100'**. The teeth **130** and **130'** press against respective the securing members **140** establishing the inside width between flange fittings **100** and **100'**. The recesses **132** and **132'** provide clearances necessary for forming the flange fitting and assembly of the vertical redirector **78**. The flange fittings **100** and **100'** do not extend beyond the first end **82** and the second end **84** of the vertical redirector base **80** or the outer edge of the receiving portion of the first end **82** or the second end **84**.

(50) For larger radius redirectors, hooks and slots similar to those shown in exemplary embodiment of a vertical redirector **478** shown in FIGS. **16C** and **16D** may be used to secure flanges to the base for additional stability.

(51) FIGS. **9B**, **9C**, and **9D** show a completely assembled vertical redirector **78**. All of the components: vertical redirector base **80**, flange fittings **100** and **100'**, eight sets of fasteners (bolts **142** and nuts **144**) comprise a vertical redirector kit for connecting two cable trays **36** (not included as part of the kit). The components within the kit may be housed within a container, such as a bag, box, bin, or the like for storage, shipment, and containing all components within the kit together until assembly and use. Cables (not shown) may be placed onto vertical redirector **78** beginning at first cable tray **36**, then across deck **86** and onto second cable tray **36'**. First cable tray **36** and second cable tray **36'** are secured to redirector base **80** using bolts **142** entering from the inside of first and second cable trays **36** and **36'**, respectively, and being fastened with nuts **144** from below the vertical redirector **78**.

(52) FIG. **10** shows a nested stack **150** of vertical redirector bases **80** of FIG. **4** for space-saving shipping and storage. Vertical redirector bases **80** can accept flange fittings **100** and two cable trays. Together vertical redirector bases **80** and flange fittings **100** of the present disclosure in quantity require considerably less volume for storage and shipping. While the vertical redirector bases **80** of FIG. **4** are shown in a nested stack **150**, the vertical redirector bases of FIGS. **7**, **13A**, **15B**, and **16A** may also form a nested stack **150** in a similar arrangement.

(53) FIG. **11** shows a nested stack **152** of flange fittings **100** of FIG. **5**, of the present disclosure for space-saving shipping or storage. Nested stack **152** uses packing space efficiently by nesting flange fittings **100**. Nesting enables the number of units stored per unit of storage volume to increase as the number of units being stored increases: A plurality of flange fittings **100** can be stored more efficiently than one fully assembled vertical redirector of prior art welded redirectors. Nesting of items means that when nested stack **150** of vertical redirector bases **80** and nested stack **152** of flange fittings **100** are put in adjacent relationship, as shown in FIG. **12**, at least part of the unused storage volume required to store the vertical redirector bases **80** or flange fittings **100** is used by the

next part, so that the percent of unused volume in a stack of parts declines as the nesting efficiency increases. While the flange fittings **100** of FIG. **5** are shown in a nested stack **152**, the flange fittings of FIGS. **8**, **13B**, **15A**, **15C**, and **16B** may also form a nested stack **152** in a similar arrangement.

(54) Another exemplary embodiment of a vertical redirector **278** is shown in FIGS. **13A**, **13B**, **14A**, **14B**, and **14C**. Vertical redirector base **280** includes a first end **282**, a second end **284**, wherein the first end **282** and the second end **284** are spaced apart from each other, a deck **286**, connecting the first end **282** and the second end **284**, and a pair of opposed tabs (**288a**, **288b**). The first end **282** and the second end **284** contain a front surface, a back surface, and an outer end. The deck **286** contains a front surface, a back surface, and a pair of opposed tabs (**288a**, **288b**) on the sides of the deck **286**. The deck **286** is preferably curved as illustrated in FIG. **13A**. An optional first transition **290** is defined by the intersection of the deck **286** and the first end **282**, and a second transition **292** is defined by the intersection of the deck **286** and the second end **284**. The first transition **290** serves as a transition portion between the deck **286** and the first end **282** that is offset from the deck **286**. The offset at the first transition **290** between deck **286** and first end **282** provides a smooth transition and receiving portion of a cable tray **36** placed adjacent the first end **282**, reducing the probability of cable damage by providing a smooth transition between the connected cable tray **36** and vertical redirector base **280**.

(55) Likewise, an optional second transition **292** serves as a transition portion between the deck **286** and the second end **284** that is offset from the deck **286**. The offset at the second transition **292** between deck **286** and second end **284** provides a smooth transition and receiving portion of a cable tray **36** placed adjacent the second end **284**, reducing the probability of cable damage by providing a smooth transition between the connected cable tray **36** and vertical redirector **278**.

(56) The tabs (**288a**, **288b**) of deck **286** engage flange fittings **200**, shown in FIGS. **14A**, **14B**, and **14C**. FIG. **13B** is a perspective view of a flange fitting **200**. The flange fitting **200** facilitate a cable redirection of about 90 degrees, for the sake of the present example. Those of ordinary skill will readily recognize that other angles for redirection are readily obtainable.

(57) Referring specifically to FIGS. **14A** and **14B**, the deck **286** of vertical redirector base **280** of FIG. **13A** may receive a flange fitting **200** of FIG. **13B** against tab **288a** and tab **288b**, as shown in FIGS. **14A** and **14B**. Flange fitting **200** includes a central portion **202**, a first end **204**, and a second end **206**. First end **204** carries a first flange **208** with at least one hole **210** formed therein. Second end **206** carries a second flange **212** with at least one hole **214**. Holes **210**, **214**, that are provided for attaching flange fitting **200** against tabs (**288a**, **288b**) of deck **286**, are preferably slotted holes to provide flexibility for connecting industrial components together.

(58) The central portion **202** of the flange fitting **200** contains an angled tab **201**. The central portion **202** is separated from the first flange **208** and the second flange **212** by a recess **232**. The central portion **202** contains a first side portion and a second side portion, wherein the first side portion extends from the recess **232** adjacent the first flange **208** and is angled away from the first flange **208**. The second side portion of the central portion **202** extends from the recess **232** adjacent the second flange **212** and is angled away from the second flange **212**. A top portion extends between the first side portion and the second side portion, forming the outer edge of the central portion **202**. As shown in FIG. **13B**, the width of the central portion **202** decreases as the central portion proceeds from the recesses **232** to the outer edge of the top portion. A hole **203** is formed within the central portion **202**.

(59) Each tab (**288a**, **288b**) contains an outer edge **234** that extends to two opposed, spaced-apart shoulders **236**. The shoulders **236** extend from an outer edge of the deck **286** to the outer edge **234** of the tab (**288a**, **288b**) and on opposite ends of the deck **286**. A first shoulder **236** is disposed adjacent the first end **282** and the second shoulder **236** is disposed adjacent the second end **284**. As illustrated in FIGS. **13A** and **14A**, the shoulders **236** extend downwardly from the outer edge of the tabs (**288a**, **288b**). A corner **238** is formed at the intersection of the shoulders **236** and the outer

edge of the deck **286**.

(60) FIGS. **14A** and **14B** show perspective views to illustrate moving flange fittings **200** into position onto deck **286** to form a vertical redirector **278** of the present disclosure. Flange fittings **200** are moved toward each side of the deck **286**, with tab **288a** centered so it can slide between first flange **208** and second flange **212** of one flange fitting **200**, and tab **288b** is centered so it can slide between first flange **208'** and second flange **212'** of the second flange fitting **200'** on the opposite side of the deck **286**. First flange **208** and second flange **212** slide under deck **286** as the outer edge of the tab **288a** contacts the inner surface of the central portion **202** of the flange fitting **200**. The shoulders **236** on opposed sides of the tab **288a** are received between the recesses **232** of the flange fitting **200**. The flange fitting **200** is positioned so that holes **294**, **296**, in first end **282** and second end **284**, respectively, are aligned with at least one hole **210** in first flange **208** or at least one hole **214** in second flange **212**, respectively, of the one flange fitting **200**. Similarly, first flange **208'** and second flange **212'** of the second flange fitting **200'** slide under deck **286** as the outer edge of the tab **288b** contacts the inner surface of the central portion **202'** of the flange fitting **200'**. The shoulders **236'** on opposed sides of the tab **288b** are received within the recesses **232'** of the flange fitting **200'**. The flange fitting **200'** is positioned so that holes **294**, **296**, in first end **282** and second end **284**, respectively, are aligned with at least one hole **210'** in first flange **208'** or at least one hole **214'** in second flange **212'** of the second flange fitting **200'**.

(61) A support member **207** is positioned adjacent the vertical redirector base **280**, as shown in FIG. **14A**. The support member **207** contains a cylindrical body portion and externally threaded cylindrical end portions. The cylindrical end portions are inserted through the hole **203** within the central portions **202**, **202'** of the first flange fitting **200** and the second flange fitting **200'**. Fastening elements **209**, such as internally threaded nuts, are engaged to externally threaded cylindrical portions for providing support and stability to the vertical redirector base **280**. The flange fitting **200** and **200'** does not extend beyond the first end **282** and the second end **284** of the vertical redirector base **280** or the outer edge of the receiving portion of the first end **282** or the second end **284**.

(62) FIGS. **14B** and **14C** show a completely assembled vertical redirector **278** according to this embodiment. All of the components: vertical redirector base **280**, flange fittings **200** and **200'**, support member **207**, two fastening elements **209**, and eight sets of fasteners (bolts **342** and nuts **344**) comprise a vertical redirector kit for connecting two cable trays **36** (not included as part of the kit). The components within the kit may be housed within a container, such as a bag, box, bin, or the like for storage, shipment, and containing all components within the kit together until assembly and use.

(63) As illustrated in FIG. **14B**, first cable tray **36** and second cable tray **36'** are secured to redirector base **280** using bolts **342** entering from the inside of first and second cable trays **36** and **36'**, respectively, and being fastened with nuts **344** from below the vertical redirector **278**.

(64) Where cable tray redirectors may be needed for a special or modified configuration in smaller quantities to satisfy a customer's specific requirement for a cable tray arrangement, the general design features of the present disclosure have an economic advantage for design and fabrication.

(65) FIGS. **15A**, **15B**, **15C**, **15D**, and **15E** show a perspective view of a narrow vertical redirector **378**. The flange fittings **300a** and **300b** within this embodiment have similar structural features as the flange fitting illustrated in FIG. **5**. In this embodiment, the deck **386** of vertical redirector base **380** may receive the flange fittings **300a** and **300b**, as shown in FIGS. **15D** and **15E**. An example of a vertical redirector base **380**, according to the present disclosure, is shown in perspective in FIG. **15B**. Vertical redirector base **380** includes a first end **382** and a second end **384**, wherein the first end **382** and the second end **384** are spaced apart from each other, and the deck **386** connects the first end **382** and the second end **384**. The first end **382** and the second end **384** contain a front surface, a back surface, and an outer end. The deck **386** contains a front surface, a back surface, and a pair of opposed tabs (**388a**, **388b**) on the sides of the deck **386**. The deck **386** is preferably

curved as illustrated in FIG. 15B. An optional first transition **390** is defined by the intersection of the deck **386** and the first end **382**, and a second transition **392** is defined by the intersection of the deck **386** and the second end **384**. The first transition **390** serves as a transition portion between the deck **386** and the first end **382** that is offset from the deck **386**. The offset at the first transition **390** between the deck **386** and the first end **382** provides a smooth transition and can receive a portion of the cable tray **40'** placed adjacent the first end **382**, reducing the probability of cable damage by providing a smooth transition between the connected cable tray **40'** and vertical redirector base **380**. (66) Likewise, an optional second transition **392** serves as a transition portion between the deck **386** and the second end **384** that is offset from the deck **386**. The offset at the second transition **392** between deck **386** and second end **384** provides a smooth transition and can receive a portion of a cable tray **40** placed adjacent the second end **384**, reducing the probability of cable damage by providing a smooth transition between the connected cable tray **40** and vertical redirector **378**. A securing member **340** extends outwardly from both of the two opposed sides of the deck **386** of the first end **382** and the second end **384**, as also illustrated in FIG. 15B.

(67) Flange fitting **300a** includes a curved portion **302a**, a first end **304a** and a second end **306a**. First end **304a** carries a first flange **308a** with at least one hole **310a** formed therein. Second end **306a** carries a second flange **312a** with at least one hole **314a** formed therein. Holes **310a** and **314a** are provided for attaching flange fitting **300a** onto the vertical redirector base **380**, are preferably slotted holes to provide flexibility for connecting industrial components together.

(68) The flange fitting **300a** contains a tooth **330a** extending outwardly from the curved portion **302a** adjacent the first flange **308a**. A recess **332a** is disposed within the curved portion **302a**, extending from the outer edge of the curved portion **302a** and into the curved portion **302a** between the tooth **330a** and the first flange **308a** and having a u-shape.

(69) Flange fitting **300b** includes a curved portion **302b**, a first end **304b** and a second end **306b**. First end **304b** carries a first flange **308b** with at least one hole **310b** formed therein. Second end **306b** carries a second flange **312b** with at least one hole **314b** formed therein. Holes **310b** and **314b** are provided for attaching flange fitting **300b** onto the vertical redirector base **380** are preferably slotted holes to provide flexibility for connecting industrial components together.

(70) Flange fitting **300b** contains a tooth **330b** extending outwardly from the curved portion **302b** adjacent the second flange **312b**. A recess **332b** is disposed within the curved portion **302b**, extending from the outer edge of the curved portion **302b** and into the curved portion **302b** between the tooth **330b** and the second flange **312b** and having a u-shape.

(71) Each tab (**388a**, **388b**) of the deck **386** contains an outer edge **334** that extends to two opposed, spaced-apart shoulders **336**. The shoulders **336** extend from an outer edge of the deck **386** to the outer edge **334** of the tab (**388a**, **388b**) and on opposite ends of the deck **386**. A first shoulder **336** is disposed adjacent the first end **382** and the second shoulder **336** is disposed adjacent the second end **384**. As illustrated in FIG. 15B, the shoulders **336** extend from the outer edge of the deck **386** at a 90° angle outwardly from the outer edge of the deck **386**. A corner **338** is formed at the intersection of the shoulders **336** and the outer edge of the deck **386**. While the shoulders **336** are adjacent the first end **382** and the second end **384**, respectively, the shoulders **336** are spaced apart from the first end **382**, first transition **390**, second end **384**, and second transition **392**, as illustrated in FIG. 15B. A securing member **340** extends outwardly from both of the two opposed sides of the first end **382** and the second end **384**, as also illustrated in FIG. 15B.

(72) FIGS. 15D and 15E show perspective views to illustrate moving flange fittings **300a** and **300b** into position onto deck **386** to form a vertical redirector **378** of the present disclosure. The front surface or receiving portion of the first end **382** is placed adjacent the back surface of a cable tray **40'**. The front surface or receiving portion of the second end **384** is placed adjacent the back surface of a second cable tray **40**. Flange fittings **300a** and **300b** are moved toward each side of the deck **386**, with tab **388a** centered so it can slide between the first flange **308a** and the second flange **312a** of flange fitting **300a**, and tab **388b** is centered so it can slide between the first flange **308b**

and second flange **312b** of the second flange fitting **300b** on the opposite side of the deck **386**. As illustrated in FIG. 15D, tab **388a** is positioned between the two teeth **330a** on the first flange fitting **300a** and the inner surface of the tab **388a** contacts the bottom **316a** of the curved portion **302a**, wherein the tab **388a** has the same radius of curvature as the bottom **316a** of the curved portion **302a** of flange fitting **300a**. Tab **388b** is positioned between the two teeth **330b** on the second flange fitting **300b** and the inner surface of the tab **388b** contacts the bottom **316b** of the curved portion **302b**, wherein the tab **388b** has the same radius of curvature as the bottom **316b** of the curved portion **302b** of the second flange fitting **300b**.

(73) The first flange **308b** and second flange **312b** of flange fitting **300b** slide over deck **386** as the tab **388b** slides over the bottom **316b** of the flange fitting **300b** and is positioned so that holes **394**, **396**, in deck **386** are aligned with at least one hole **310b** in first flange **308b** or at least one hole **314b** in second flange **312b**, respectively, of the flange fitting **300b**. The first flange **308b** and the second flange **312b** of flange fitting **300b** have a different configuration than the first flange **308a** and the second flange **312a** of flange fitting **300a**. The first flange **308b** and the second flange **312b** of flange fitting **300b** are offset, meaning the first flange **308b** and the second flange **312b** have an increased elevation with respect to the curved portion **302b** compared to the elevation of first flange **308a** and second flange **312a** with respect to the curved portion **302a** of flange fitting **300a**. The first flange **312a** and the second flange **312b** have an increased elevation of one material thickness or the thickness of the first flange **308b** and the second flange **312b** compared to the first flange **308b** and second flange **312b**. This increased elevation allows the first flange **308b** to slide overtop second flange **312a** and second flange **312b** to slide overtop first flange **308a**, as illustrated in FIG. 15E. As first flange **308b** slides overtop second flange **312a** and second flange **312b** slides overtop first flange **308a**, the tab **388a** slides over the bottom **316a** of the flange fitting **300a** and is positioned so that holes **394**, **396**, in first end **382** and second end **384**, respectively, are aligned with at least one hole **310a** in first flange **308a** or at least one hole **314a** in second flange **312a**, respectively, of the flange fitting **300a**. The teeth **330a** and **330b** press against respective securing members **340** establishing the inside width between flange fittings **300a** and **300b**. The recesses **332a** and **332b** provide clearances necessary for forming the flange fittings **300a** and **300b** and assembly of the vertical redirector **378**. The flange fittings **300a** and **300b** do not extend beyond the first end **382** and the second end **384** of the vertical redirector base **380** or the outer edge of the receiving portion of the first end **382** or the second end **384**.

(74) In an alternate configuration, **300a** and **300b** are identical flanges. The positioning of the flanges, how they fit around deck **386**, and how they assemble the vertical redirector **378** would be the same as described for those that are not identical.

(75) FIG. 15E shows a completely assembled vertical redirector **378**, according to this embodiment. All of the components: vertical redirector base **380**, flange fittings **300a** and **300b**, four sets of fasteners (bolts **342** and nuts **344**) comprise a vertical redirector kit for connecting two cable trays **40** and **40'** (not included as part of the kit). The components within the kit may be housed within a container, such as a bag, box, bin, or the like for storage, shipment, and containing all components within the kit together until assembly and use.

(76) Cables (not shown) may be placed onto vertical redirector **378** beginning at first cable tray **40**, then across deck **386** and onto second cable tray **40'**. First cable tray **40** and second cable tray **40'** are secured to redirector base **380** using bolts **342** entering from the inside of first and second cable trays **40** and **40'**, respectively, and being fastened with nuts **344** from below the vertical redirector **378** as illustrated in FIGS. 15D and 15E.

(77) Another exemplary vertical redirector base **480**, according to the present disclosure, is shown in perspective in FIG. 16A. Vertical redirector base **480** includes a first end **482**, a second end **484**, wherein the first end **482** and the second end **484** are spaced apart from each other, a deck **486**, connecting the first end **482** and the second end **484**, and a pair of opposed tabs (**488a**, **488b**). The first end **482** and the second end **484** contain a front surface, a back surface, and an outer end. The

deck **486** contains a front surface, a back surface, and a pair of opposed tabs (**488a**, **488b**) on the sides of the deck **486**. The deck **486** is preferably curved as illustrated in FIG. **16A**. Each tab (**488a**, **488b**) contains an outer edge that extends to two opposed, spaced-apart shoulders **436**. The shoulders **436** extend from an outer edge of the deck **486** to the outer edge of the tab (**488a**, **488b**) and on opposite ends of the deck **486**. A first shoulder **436** is disposed adjacent the first end **482** and the second shoulder **436** is disposed adjacent the second end **484**. As illustrated in FIG. **16A**, the shoulders **436** extend from the outer edge of the deck **486** at a 90° angle outwardly from the outer edge of the deck **486**. A corner **438** is formed at the intersection of the shoulders **436** and the outer edge of the deck **486**. At least one hook **500** is disposed on tab **488a**, extending outwardly from the tab **488a**. The hook **500** contains a base portion that extends outwardly from the deck **486**, extending outwardly from the tab **488a** of the deck **486**, and a perpendicularly extending elongate portion from the base that is spaced-apart from the deck **486** and tab **488a**, creating a void between the elongate portion and the deck **486** and tab **488a**. As illustrated in FIG. **16A**, two hooks **500** are disposed on tab **488a**.

(78) Likewise, at least one hook **502** is disposed on tab **488b**, extending outwardly from the tab **488b**. The hook **502** contains a base portion that extends outwardly from the deck **486**, extending outwardly from the tab **488b** of the deck **486**, and a perpendicularly extending elongate portion from the base that is spaced-apart from the deck **486** and tab **488b**, creating a void between the elongate portion and the deck **486** and tab **488b**. As illustrated in FIG. **16A**, two hooks **502** are disposed on tab **488b**, wherein the hooks **502** face in an opposite direction than the hooks **500**. In other words, the elongate portion of the hooks **502** extends perpendicularly outwardly in the opposite direction than the elongate portion of the hooks **500**. As illustrated in FIG. **16A**, the elongate portion of the hooks **500** extend perpendicularly outward towards the first end **482** of the deck, while the elongate portion of the hooks **502** extend perpendicularly outward towards the second end **484** of the deck **486**. Hooks **500** and **502** assist in the alignment of the redirector components and secure the final assembly. Hooks **500** and **502** and deck **486** are each curved at the same radius for a matched fit at assembly. Larger redirectors with larger radius bend may require additional hooks. While two hooks (**500**, **502**) are illustrated on each tab (**488a**, **488b**) of the deck **486**, the base **480** may include two or more hooks (**500**, **502**) on each tab (**488a**, **488b**) or may contain a plurality of hooks (**500**, **502**) on each tab (**488a**, **488b**).

(79) The deck **486** of vertical redirector base **480** of FIG. **16A** may receive two curved flange fittings **400** having a structure illustrated in FIG. **16B**. Flange fitting **400** includes a curved portion **402** with a bottom **416**, a first end **404**, and a second end **406**. First end **404** carries a first flange **408** with at least one hole **410** formed therein. Second end **406** carries a second flange **412** with at least one hole **414** formed therein. Holes **410**, **414** that are provided for attaching flange fitting **400** are preferably slotted holes to provide flexibility for connecting industrial components together. At least one slot **504** is disposed within the curved portion **402** of the flange fitting **400** and preferably positioned in close proximity to the bottom **416** of the curved portion **402**. As illustrated in FIG. **16B**, the flange fitting **400** contains two spaced-apart slots disposed within the curved portion **402** of the flange fitting **400** in close proximity to the bottom **416** of the curved portion **402**. Slots **504** and deck **486** are each curved at the same radius for a matched fit at assembly. Larger redirectors with a larger radius bend may require additional slots. The flange fitting **400** may contain one slot **504**, two or more slots **504**, or a plurality of slots **504**. The number of slots **504** preferably corresponds to the number of hooks **500**. A recess **432** is disposed within the curved portion **402**. The flange fitting **400** facilitates a cable redirection of about 90 degrees, for the sake of the present example. Those of ordinary skill will readily recognize that other angles for redirection are readily obtainable.

(80) FIG. **16C** shows a perspective view to illustrate moving flange fittings **400** and **400'** into position onto deck **486** to form a vertical redirector **478** of the present disclosure. Flange fittings **400** and **400'** are moved toward each side of the deck **486**, wherein each slot **504** is centered with a

hook **500** so the slot **504** can slide over the hook **500**, resulting in the elongate portion of the hook **500** sliding through the slot **504** and positioned on the outer side of the flange fitting **400** with a portion of the slot **504** positioned within the void of the hook **500**. Hooks **500** and **502**, deck **486**, and slots **504** and **504'** are each curved at the same radius for a matched fit at assembly. The flange fitting **400** is slid in the direction opposite the direction the elongate portion of the hook **500** extending until an end of the slot **504** contacts the base of the hook **500**, preventing further movement of the flange fitting **400**, creating a stable and secure connection between the flange fitting **400** and vertical redirector base **480**. Similarly, slot **504'** is centered with a hook **502** so the slot **504'** can slide over the hook **502**, resulting in the elongate portion of the hook **502** sliding through the slot **504'** and positioned on the outer side of the flange fitting **400'** with a portion of the slot **504'** positioned within the void of the hook **502**. The flange fitting **400'** is slid in the direction opposite the direction the elongate portion of the hook **502** extending until an end of the slot **504'** contacts the base of the hook **502**, preventing further movement of the flange fitting **400'**, creating a stable and secure connection between the flange fitting **400'** and vertical redirector base **480**. The flange fitting **400'** may contain one slot **504'**, two or more slots **504'**, or a plurality of slots **504'**. The number of slots **504'** preferably corresponds to the number of hooks **504'**.

(81) First flange **408** and second flange **412** slide under deck **486** as slots **504** slide over hooks **500** of the flange fitting **400**. The flange fitting **400** is positioned so that holes **494**, **496**, in first end **482** and second end **484**, respectively, are aligned with at least one hole **410** in first flange **408** or at least one hole **414** in second flange **412**, respectively, of flange fitting **400**. Similarly, first flange **408'** and second flange **412'** of the second flange fitting **400'** slide under deck **486** as slots **504'** of the flange fitting **400'** slide over hooks **502**. The flange fitting **400'** is positioned so that holes **494**, **496**, in first end **482** and second end **484**, respectively are aligned with at least one hole **414'** in first flange **412'** or at least one hole **410'** in second flange **408'**, respectively, of the second flange fitting **400'**.

(82) FIG. **16D** shows a completely assembled vertical redirector **478**. All of the components: vertical redirector base **480**, flange fittings **400** and **400'**, eight sets of fasteners (bolts **442** and nuts **444**) comprise a vertical redirector kit for connecting two cable trays **36** and **36'** (not included as part of the kit). The components within the kit may be housed within a container, such as a bag, box, bin, or the like for storage, shipment, and containing all components within the kit together until assembly and use.

(83) It should be noted that in FIG. **16C**, a total of eight bolts **442** and nuts **444** are needed to hold two cable trays **36**, **36'** within the non-welded cable tray redirector. FIG. **16D** shows the same number, eight bolts **442** and nuts **444** used to hold two cable trays **36**, **36'** to the vertical redirector **478** without any welding needed to complete the vertical redirector components or assembly. The cost and time to weld cable tray redirectors has been completely avoided. Cables (not shown) may be placed onto vertical redirector **478** beginning at first cable tray **36**, then across deck **486** and onto second cable tray **36'**. First cable tray **36** and second cable tray **36'** are secured to redirector base **480** using bolts **442** entering from the inside of first and second cable trays **36** and **36'**, respectively, and being fastened with nuts **444** from below the vertical redirector **478**.

(84) Connecting fasteners may be provided together with cable tray redirectors or in certain situations may be provided by others according to customer's requirements. Connecting fasteners with low profile heads provide greater capacity inside the tray system. Additional fasteners may be provided for tray runs for mounting or installing the cable tray run or to connect other components to the cable tray run.

(85) Connecting fasteners are preferably bolts and other "tightenable" fasteners, because they are generally better able to maintain the electrical continuity needed to meet industrial electrical codes and regulations.

(86) If bolts are used for assembling tray redirectors and the tray system, nuts, locking nuts, or flanged locking nuts may be used with or without washers, to create contact pressure and contact

surface area, between the redirector and the tray surfaces for electrical continuity, which may be useful to meet industrial electrical codes and regulations.

(87) The fasteners that are used to assemble a cable tray redirector, according to the present disclosure, may be the same fasteners that connect cable tray ends to the cable tray redirector.

(88) Where redirecting connectors have larger bend radii or longer wall lengths the curved or straight diagonal flange fittings may be reinforced with ribs, fasteners, brackets or the like for rigidity of the curved or diagonal flange within the overall assembly.

(89) The elimination of welding, heating or stressing metal in cable tray components reduces cost and increases product desirability for use in safety-related products or products used by the nuclear industry or facilities where there are seismic restrictions.

(90) Since many cable trays have multiple rows of parallel holes and most cable redirectors connect two tray sections, the quantities of horizontal 90° curved flange fittings that require offset or staggered flanges is normally relatively small. Therefore, curved flange or diagonal straight flange fittings are generally identical throughout the redirectors of a cable tray run where the cable tray is of the same size and configuration and the degree and radius of redirections does not vary. Where more than two intersecting, narrow cable tray sections having a single row of holes for connecting a tray to redirectors intersect, a combination of higher and lower flanges can be used to construct a “T” or cross redirector.

Claims

1. A vertical redirector for use with at least two cable trays, comprising: (a) a vertical redirector base having a first end and a second end spaced-apart from each other, a deck connecting the first end and the second end, wherein the first end contains a first receiving portion and the second end contains a second receiving portion, the deck contains a first side and an opposed second side with a first tab extending from the first side of the deck and a second tab extending from the second side of the deck; (b) a first flange fitting having a first end and a second end, the first end carries a first flange with at least one hole formed in the first flange being registrable with at least one hole formed on the first receiving portion, the second end carries a second flange with at least one hole formed in the second flange being registrable with at least one hole formed on the second receiving portion, a curved portion extending between the first end and the second end of the first flange fitting; and (c) a second flange fitting having a first end and a second end, the first end carries a first flange with at least one hole formed in the first flange being registrable with at least one hole formed on the first receiving portion, the second end carries a second flange with at least one hole formed in the second flange being registrable with at least one hole formed on the second receiving portion, a curved portion extending between the first end and the second end of the second flange fitting.

2. The vertical redirector of claim 1, further comprising at least one fastener securing the first flange fitting and the second flange fitting to the vertical redirector base.

3. The vertical redirector of claim 1, wherein the curved portion of the flange fitting is c-shaped.

4. The vertical redirector of claim 1, wherein the curved portion of the flange fitting is flat.

5. The vertical redirector of claim 1, wherein the first flange of the first flange fitting and the second flange fitting is joined to the first receiving portion of the vertical redirector base and the second flange of the first flange fitting and the second flange fitting is joined to the second receiving portion of the vertical redirector base.

6. The vertical redirector of claim 1, wherein the first end of the first flange fitting and the first end of the second flange fitting do not extend beyond the end of the first end of the vertical redirector base.

7. The vertical redirector of claim 1, wherein the vertical redirector base is configured to nest with a second vertical redirector base.

8. The vertical redirector of claim 1, wherein the flange fitting is configured to nest with another flange fitting.

9. A vertical redirector for use with at least two cable trays, comprising: (a) a vertical redirector base having a first end and a second end spaced-apart from each other, a deck connecting the first end and the second end, wherein the first end contains a first receiving portion and the second end contains a second receiving portion, the deck contains a first side and an opposed second side with a first tab extending from the first side of the deck and a second tab extending from the second side of the deck, a first notch is disposed within the first tab and a second notch is disposed within the second tab; (b) a first flange fitting having a first end and a second end, the first end carries a first flange with at least one hole formed in the first flange being registrable with at least one hole formed on the first receiving portion, the second end carries a second flange with at least one hole formed in the second flange being registrable with at least one hole formed on the second receiving portion, a curved portion extending between the first end and the second end of the first flange fitting; and (c) a second flange fitting having a first end and a second end, the first end carries a first flange with at least one hole formed in the first flange being registrable with at least one hole formed on the first receiving portion, the second end carries a second flange with at least one hole formed in the second flange being registrable with at least one hole formed on the second receiving portion, a curved portion extending between the first end and the second end of the second flange fitting.

10. The vertical redirector of claim 9, further comprising at least one fastener securing the first flange fitting and the second flange fitting to the vertical redirector base.

11. The vertical redirector of claim 9, wherein the curved portion of the flange fitting is c-shaped.

12. The vertical redirector of claim 9, wherein the curved portion of the flange fitting is flat.

13. The vertical redirector of claim 9, wherein the first flange of the first flange fitting and the second flange fitting is joined to the first receiving portion of the vertical redirector base and the second flange of the first flange fitting and the second flange fitting is joined to the second receiving portion of the vertical redirector base.

14. The vertical redirector of claim 9, wherein the first end of the first flange fitting and the first end of the second flange fitting do not extend beyond the end of the first end of the vertical redirector base.

15. The vertical redirector of claim 9, wherein the vertical redirector base is configured to nest with a second vertical redirector base.

16. The vertical redirector of claim 9, wherein the flange fitting is configured to nest with another flange fitting.

17. A vertical redirector for use with at least two cable trays, comprising: (a) a vertical redirector base having a first end and a second end spaced-apart from each other, a deck connecting the first end and the second end, wherein the first end contains a first receiving portion and the second end contains a second receiving portion, the deck contains a first side and an opposed second side with a first tab extending from the first side of the deck and a second tab extending from the second side of the deck, a first notch is disposed within the first tab and a second notch is disposed within the second tab; (b) a first flange fitting having a first end and a second end, the first end carries a first flange with at least one hole formed in the first flange being registrable with at least one hole formed on the first receiving portion, the second end carries a second flange with at least one hole formed in the second flange being registrable with at least one hole formed on the second receiving portion, a central portion extending between the first end and the second end of the first flange fitting and disposed between the first flange and the second flange containing at least one hole; and (c) a second flange fitting having a first end and a second end, the first end carries a first flange with at least one hole formed in the first flange being registrable with at least one hole formed on the first receiving portion, the second end carries a second flange with at least one hole formed in the second flange being registrable with at least one hole formed on the second receiving portion, a

central portion extending between the first end and the second end of the second flange fitting and disposed between the first flange and the second flange containing at least one hole.

18. The vertical redirector of claim 17, further comprising a support member containing a first externally threaded end that is received within the hole of the central portion of the first flange fitting and a second externally threaded end that is received within the hole of the central portion of the second flange fitting.

19. The vertical redirector of claim 17, wherein the first flange fitting does not extend beyond an outer edge of the receiving portion of the first end or the second end of the vertical redirector base and the second flange fitting does not extend beyond the outer edge of the receiving portion of the first end or the second end of the vertical redirector base.

20. The vertical redirector of claim 17, wherein the first flange fitting is configured to nest with the second flange fitting.
